



A New Declining Phase Precursor and an Early Prediction of Cycle 26 Maximum

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Abstract

The overall level of solar activity as seen in the sunspot number (SSN) record paces both the state of the ionosphere and the amplitude and occurrence rates of extreme space weather events. SSN predictions on solar cycle timescales are problematic given the variability in cycle amplitude and duration. The Hilbert transform of 13 month smoothed SSN since 1749 is used to construct a uniform clock for the Schwabe solar cycle, which establishes a clear switch-on and -off of geomagnetic activity seen at Earth and correlates with solar cycle morphology. Cycle SSN maxima are shown to correlate with the SSN at the switch-off of the preceding cycle. The maxima of solar cycles 1–25 are hindcasted with ~ 7 yr lead time with good fidelity by this method with coefficient of determination $r^2 > 0.7$ and Pearson correlation $\rho > 0.8$. Modeling the current SSN forward to the switch-off of cycle 25 gives an early prediction of a weak-to-moderate upcoming cycle 26 maximum, of intensity less than, or of order, cycle 25. A more precise prediction will be possible in 2.5–3.5 yr when the switch-off of cycle 25 will occur.

Unified Astronomy Thesaurus concepts: [Solar cycle \(1487\)](#); [Sunspot number \(1652\)](#)

1. Introduction

Solar and stellar dynamos are a long-standing unsolved problem in physics, and as a consequence, solar cycle variability is as yet far from understood (D. H. Hathaway 2015; P. Charbonneau & D. Sokoloff 2023). Long-term sunspot number (SSN) observations reveal an approximately 11 yr Schwabe solar cycle of activity, but no two cycles have the same amplitude or duration (A. Norton et al. 2023). Predicting the strength of the upcoming solar cycle is therefore challenging, but it can be critical for technological infrastructure. Both the risk of extreme space weather activity (K. Tsubouchi & Y. Omura 2007; E. K. J. Kilpua et al. 2015; S. C. Chapman et al. 2020; A. Bergin et al. 2022) and overall ionospheric conditions (M. Lockwood et al. 2016; M. A. Caffolla et al. 2025) correlate with the SSN. Space weather storms can disrupt power distribution, communication, aviation and satellites, and a wide range of systems that depend upon global position, navigation, and timing, with potentially significant societal impact and financial loss (E. J. Oughton et al. 2017; M. Hapgood 2019). Unexpectedly high ionospheric drag, as has occurred during the current solar maximum, can compromise satellite constellations in low Earth orbit (W. S. Shambaugh 2025; X. Wang et al. 2025). Planning and mitigation of risk then require advance knowledge of the SSN both in the form of forecasts months to years ahead and prediction of the strength of the next solar cycle focusing on the cycle SSN maximum.

There has been considerable effort to predict upcoming solar cycle maxima (D. H. Hathaway 2015; W. D. Pesnell 2020; K. Petrovay 2020; D. Nandy 2021). A range of methodologies have been employed (K. Petrovay 2020), resulting in a broad range of predicted values. The underpinning methods include physical (surface flux transport and dynamo) models (P. Bhowmik & D. Nandy 2018; P. Bhowmik et al. 2023), time series

analyses (R. N. Bracewell 1986) including nonlinear systems approaches (e.g., K. Gerçeker et al. 2025 and references therein), and neural networks and machine learning approaches (e.g., Y. Dang et al. 2022 and references therein).

There is a long-standing interest in predictors of the upcoming cycle maximum from properties of the SSN record of preceding cycles (D. H. Hathaway 2015), such as the Gnevishhev–Ohl rule, cycle rise time (the Waldemier effect), and preceding cycle length (D. H. Hathaway et al. 1994, 2002 and references therein; also S. W. McIntosh et al. 2020).

One approach is that of cycle precursors, that information about the strength of an upcoming Schwabe cycle is contained in the decline of the preceding cycle. A pair of Schwabe cycles forms a single extended (R. P. Wilson et al. 1988; E. W. Cliver 2014) Hale cycle of approximately 22 yr duration, during which the Sun completes two polarity reversals. The extended cycle can be seen in observed features of coronal morphology (S. W. McIntosh & R. J. Leamon 2014; S. W. McIntosh et al. 2014; S. W. McIntosh et al. 2019) and in the time correlation (H. H. Sargent 1985; S. C. Chapman et al. 2021) of the *aa* index, the longest continuous index of geomagnetic activity at Earth. The polar magnetic fields during the declining phase of the cycle “seed” the toroidal fields that drive sunspot activity of the next cycle (R. Cameron & M. Schüssler 2015), offering the basis of a prediction (K. H. Schatten et al. 1978; K. Schatten et al. 1996; L. Upton & D. H. Hathaway 2014). Models suggest that the magnetic topology of active regions (ARs) in the declining phase can have a substantial impact on the polar field (S. Pal et al. 2023). Observations over the last four cycles support a correlation between the decay rate of a sunspot cycle and the amplitude of the following sunspot cycle (P. Jaswal et al. 2024). We would then expect correlation between the next cycle maximum and indicators of the global solar state in the preceding declining phase and minimum. Conditions at minimum used for predictions then range from indicators of the solar polar fields to geomagnetic disturbances at Earth (L. A. Upton & D. H. Hathaway 2023); Feynman’s interplanetary component of the *aa* index is a predictor of the



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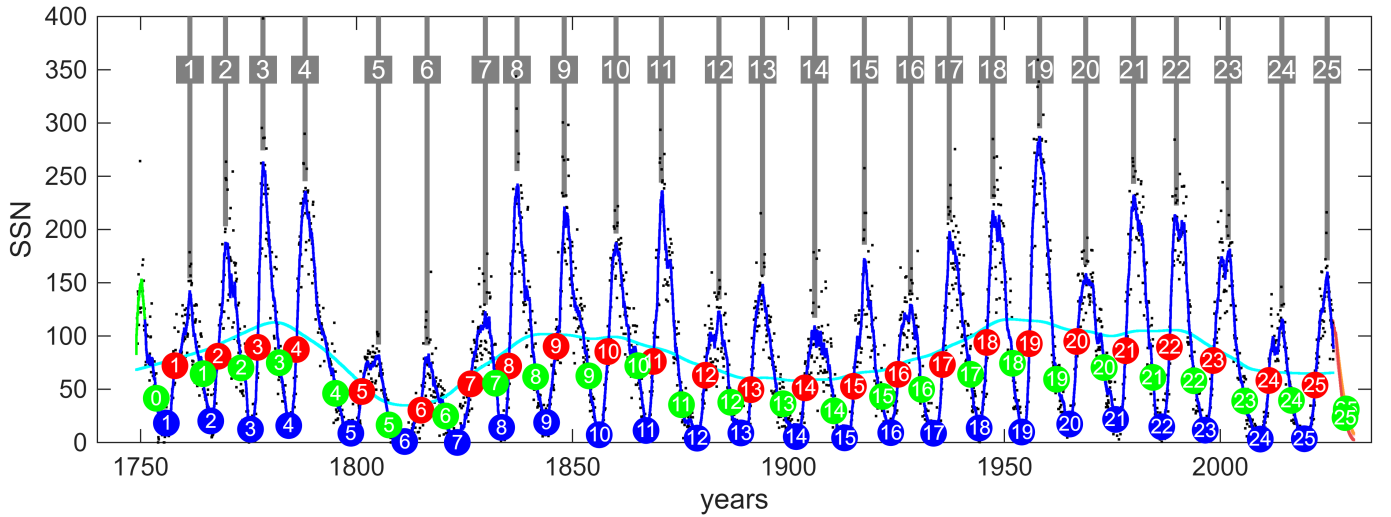


Figure 1. Analysis of the SSN record. The figure plots the last 25 cycles of monthly SSN (black), 13 month smoothed SSN (dark blue), and 40 yr trend (light blue). Each SILSO cycle maximum is indicated by gray vertical lines with the cycle number indicated. Colored numbered circles indicate each cycle minimum (blue), switch-off (green), and switch-on (red) at times determined by Hilbert transform of the SSN. The numbering convention is that a cycle maximum follows a cycle minimum, so that the N th cycle minimum occurs first, followed by the N th cycle switch-on, then maximum, then N th cycle switch-off. The Hilbert transform is performed over a whole number of cycles, so that the analysis of the SSN record discards the first $\sim 1/4$ cycle indicated in green. The projections used to extend the SSN record are indicated in orange (projection 1: NOAA), red (projection 2: S. W. McIntosh et al. (2023)), and pink (projection 3: NMSC; see also Figure 10, a zoom of the last few cycles).

strength of the following solar cycle (J. Feynman 1982; D. H. Hathaway & R. M. Wilson 2006). Comprehensive observations of the solar state are available for the last few solar cycles, providing detailed observations for a small ensemble of cycle maximum predictions. Predictions based on SSN precursors are complementary as they provide a larger ensemble with which to hindcast and obtain uncertainty estimates.

This paper identifies a new SSN precursor that linearly correlates with the maximum of the upcoming cycle. The Hilbert transform of the SSN time series can be used to map each irregular duration cycle onto a regular interval in phase (a solar cycle clock; S. C. Chapman et al. 2020), and this provides the timings of the active and quiet interval of each cycle. It is shown that the SSN at the time of the active-to-quiet switch-off of activity correlates well with the following SSN maximum, providing a method for predicting the upcoming cycle maximum on a ~ 7 yr time horizon. For cycle 26 maximum, this time horizon can be extended back to the present by modeling the SSN an additional 2.5–3.5 yr forward to the switch-off, a prediction that can be updated to achieve higher accuracy when the switch-off is reached. Timings from the solar cycle clock define the duty cycle of the SSN, revealing distinct behavior for small and large cycles with implications for solar cycle SSN projections that rely upon a typical or average solar cycle shape.

2. Solar Cycle Activity Timings

Figure 1 plots the solar SSN record that is analyzed in this paper. The SILSO v2.0 monthly SSN records (black dots) are available since 1749 (F. Clette & L. Lefèvre 2015). The 13 month smoothed SSN (dark blue) and 40 yr smoothed SSN (light blue), which tracks the Gleissberg cycle (W. Gleissberg 1967; S. C. Chapman et al. 2021), form the basis of our analysis. The 13 month smoothed SSN is routinely used to identify the solar cycle maxima and minima (hereafter SILSO maxima and minima; F. Clette & L. Lefèvre 2015).

Normalizations of the SSN Schwabe cycle onto a uniform time base have included using the sawtooth pattern in average modulus sunspot latitude (M. J. Owens et al. 2011) and collapse of the SSN profile onto a single function (M. J. Owens et al. 2011; D. H. Hathaway 2015). Here, the Hilbert transform of the SSN record is used to map each nonuniform duration solar cycle onto a uniform interval 2π of analytic phase, providing a regular solar cycle “clock” (S. C. Chapman et al. 2020; S. C. Chapman & T. Dudok de Wit 2024). Once the SSN is mapped from time to phase, the quiet interval in the solar cycle becomes well defined in terms of the occurrence rates of both solar flares and extreme geomagnetic activity at Earth (S. C. Chapman et al. 2020). In terms of analytic phase, the cycle quiet interval can be approximately centered on the average minimum, that is, the average of the phases of the SILSO cycle minima identified in the 13 month smoothed SSN record of the last 25 solar cycles. The quiet interval spans $\sim 4\pi/5$ in phase and is bracketed by a switch-off and -on of activity at phase $\pm 2\pi/5$ in advance of and following the average minimum. This quiet interval would have a duration of 4.4 yr in an exactly 11 yr cycle but is nonuniform in time in the SSN record. Mapping from analytic phase to time gives the occurrence times, shown in Figure 1, of each cycle minimum (blue circles), switch-on (red circles), and switch-off (green circles).

The procedure for mapping between time and analytic phase (see Appendix A for details) is shown in Figure 2, which plots (a) the 13 month smoothed SSN from which a 40 yr trend has been subtracted and (b) this signal extended forward in time with a prediction. The SSN record on solar cycle scales can be considered as an oscillator with irregular amplitude and period. The Hilbert transform $H(t)$ of this signal $S(t)$ is used to construct the analytic signal (D. Gabor 1946; B. Boashash 1992) $A(t)\exp[i\phi(t)] = S(t) + iH(t)$ with time-varying amplitude $A(t)$ and phase $\phi(t)$. Each solar cycle then maps to a uniform interval of 2π in phase (S. C. Chapman et al. 2020; S. C. Chapman & T. Dudok de Wit 2024), creating a solar cycle “clock” with uniform cycle lengths. Panel (c) plots the

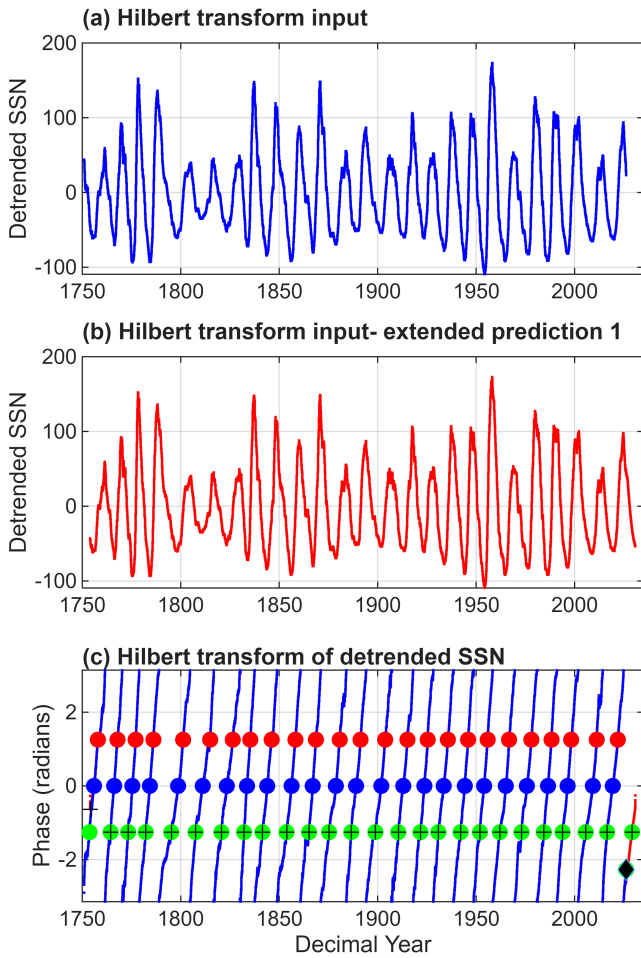


Figure 2. Constructing the solar clock and identifying the switch-on and -off of the solar cycle. The figure shows the procedure for performing the Hilbert transform of the SSN record. (a) 13 month running mean (blue) of the SILSO SSN record from which the 40 yr flowless slow trend has been subtracted. (b) SSN 13 month running mean, extended with the current NOAA projection, again with the slow trend subtracted. In both cases the signal has been truncated at the start to give a whole number of cycles as input into the Hilbert transform. (c) Hilbert transform analytic phase from (a, blue) overplotted on top of that from (b, red). Zero phase (blue circles) is set at the average phase of the SILSO minima of cycles 1–25. The switch-off (green circles) and -on (red circles) occur at times at which the phases are $\pm 2\pi/5$ either side of the averaged minimum. These are all obtained from the analytic phase of the SSN data (a), except the last switch-off, which is obtained from the analytic phase of the projection-extended SSN (b). The black crosses mark the switch-off phases/times obtained from (b). The black diamond marks the phase at the current time (at time of writing, 2026 May 1).

phase $\phi(t)$ versus time obtained for the prediction-extended SSN (red) overplotted with that obtained for the SSN (blue). The absolute signal phase is arbitrary, and here it is set to zero at the average of the phases of the SILSO minima of cycles 1–25 (blue circles). Each cycle's switch-off (green circles) and -on (red circles) are then set at phases $\pm 2\pi/5$ either side of zero. Inverting $\phi(t)$ at these phases then gives the occurrence times of the switch-on and -off for each cycle (plotted in Figure 1). The black diamond marks the phase and time of the most recent monthly SSN record from SILSO at the time of writing.

The switch-off timings will be used to hindcast the maxima of cycles 1–25. Prediction of cycle 26 maximum will require forward projection of the SSN (Figures 1 and 10, orange, red and pink lines) past the switch-off of cycle

25. Averaging over cycles 1–24, the prediction horizon for these hindcasts, that is, the time between the switch-off and the next SILSO cycle maximum, is 6.9 yr (standard deviation 1.4 yr).

3. Cycle Progression

The solar clock also orders the butterfly pattern of sunspot latitudes (S. C. Chapman & T. Dudok de Wit 2024), and this is shown in Figure 3. The switch-on and -off are indicated on Figure 3 by red and green lines respectively. The AR areas' record reports the total sunspot area (in units of millionths of a hemisphere) found in 50 latitude bins distributed uniformly in $\sin(\text{latitude})$ per Carrington rotation since 1874. Figure 3 plots the AR centroid latitudes for each Carrington rotation for each hemisphere versus the analytic phase at the time of the midpoint of each Carrington rotation, obtained from the Hilbert transform of the 13 month smoothed, 40 yr trend subtracted SSN (see Appendices A, B). The ARs appear in a highly intermittent manner, but when their overall latitudinal path is tracked, it forms part of an extended cycle of activity (R. P. Wilson et al. 1988; E. W. Cliver 2014) that commences at high latitudes and terminates at the equator (S. W. McIntosh & R. J. Leamon 2014). The extended cycle latitudinal bands can be identified in multiple features of activity (E. W. Cliver 2014; S. W. McIntosh et al. 2023 and references therein). The yellow lines on Figure 3 are a simple model for the track of the extended cycle. The simple extended cycle model is straight lines passing through coordinates (Hilbert phase, latitude) of $(2n + 1)\pi, \pm \pi/12$ and $(2n\pi + 2\pi/5, 0)$. The AR centroids and model are repeated to show an average Hale (22 yr) cycle. This model is a simplification of empirical linear fitting of the AR centroid latitudes as a function of Hilbert phase (Appendix B).

The model intersects zero latitude at the switch-on, which roughly coincides (S. C. Chapman et al. 2020; S. C. Chapman & T. Dudok de Wit 2024) with the average of terminator times (S. W. McIntosh et al. 2019) that have been directly identified as the time when EUV bright points annihilate at the equator. The terminator has been identified as the onset of the new active phase of the cycle in multiple observations of coronal magnetic activity (S. W. McIntosh & R. J. Leamon 2014; S. W. McIntosh et al. 2014; S. W. McIntosh et al. 2019). The switch-off coincides with a transition in the character of geomagnetic activity seen in the *aa* index (S. C. Chapman et al. 2020; S. C. Chapman & T. Dudok de Wit 2024). Before the switch-off, geomagnetic storms are more extreme, whereas after the switch-off, they are generally more moderate, and their occurrence coincides with time correlation at the 27 day period of solar rotation (S. C. Chapman & T. Dudok de Wit 2024). This is consistent with transition from CME to high-speed stream-driven geomagnetic storms. The switch-off also occurs after the bulk of sunspot ARs are observed at latitudes within 15° ($\pi/12$, indicated by horizontal blue lines in Figure 3) of the solar equator, where solar differential rotation is reduced (J. G. Beck 2000; D. H. Hathaway et al. 2022; S. C. Chapman & T. Dudok de Wit 2024), again favoring high-speed streams over shear-driven CMEs. In this sense, the switch-off identifies the timing of the declining phase for each solar cycle.

The progression of the current solar cycle 25 is also shown in Figure 3, where orange circles overplot the AR centroids for the last 80 Carrington rotations. We have yet to reach the

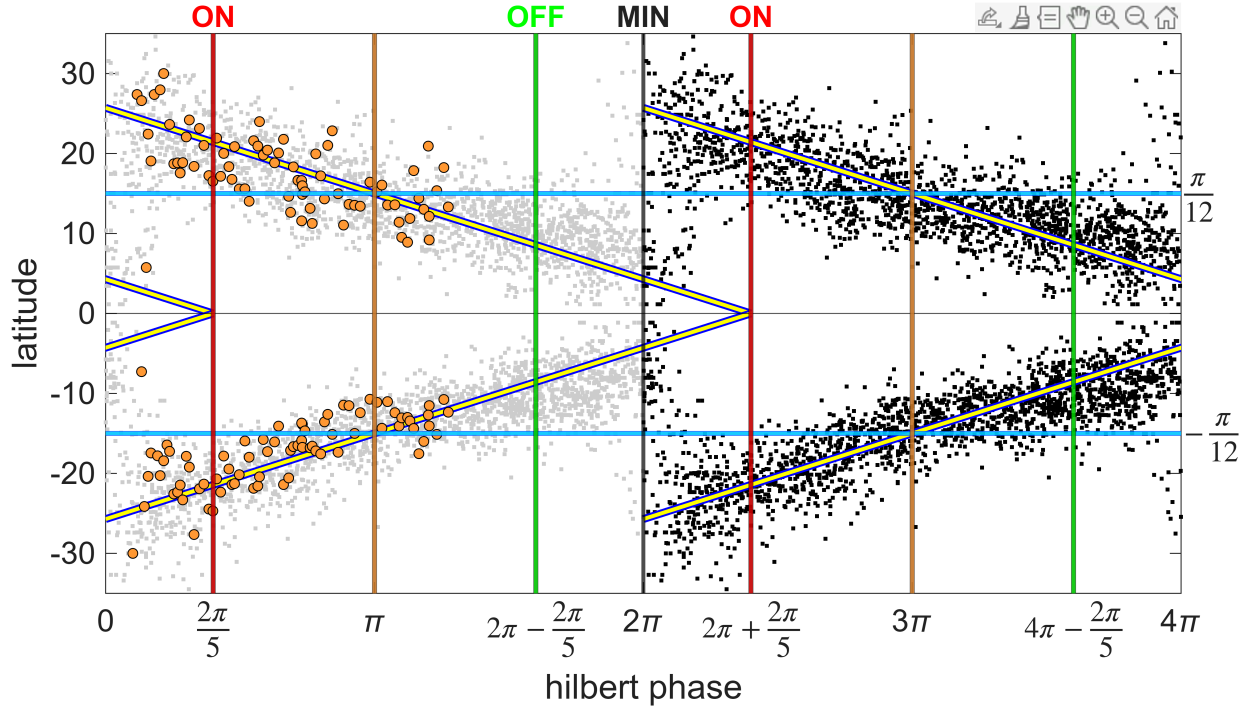


Figure 3. Extended cycle and progression of cycle 25. Sunspot AR centroid latitudes are plotted versus the Hilbert phase at the midpoint of each Carrington rotation. The AR centroids are overplotted from 1874 (approximately cycles 11–25, black points). Orange circles overplot the AR centroids for the last 80 Carrington rotations. The AR centroids and model are repeated to show an aggregate Hale (22 yr) cycle. The switch-on and -off are indicated by vertical red and green lines, respectively; the plot is centered on the average minimum, which is at zero (2π) phase (black line). The simple extended cycle model is straight lines (yellow lines) passing through coordinates (Hilbert phase, latitude) of $((2n + 1)\pi, \pm \pi/12)$ and $(2n\pi + 2\pi/5, 0)$. These coordinates are the intersections of the blue and orange lines and the switch-on at zero latitude, respectively.

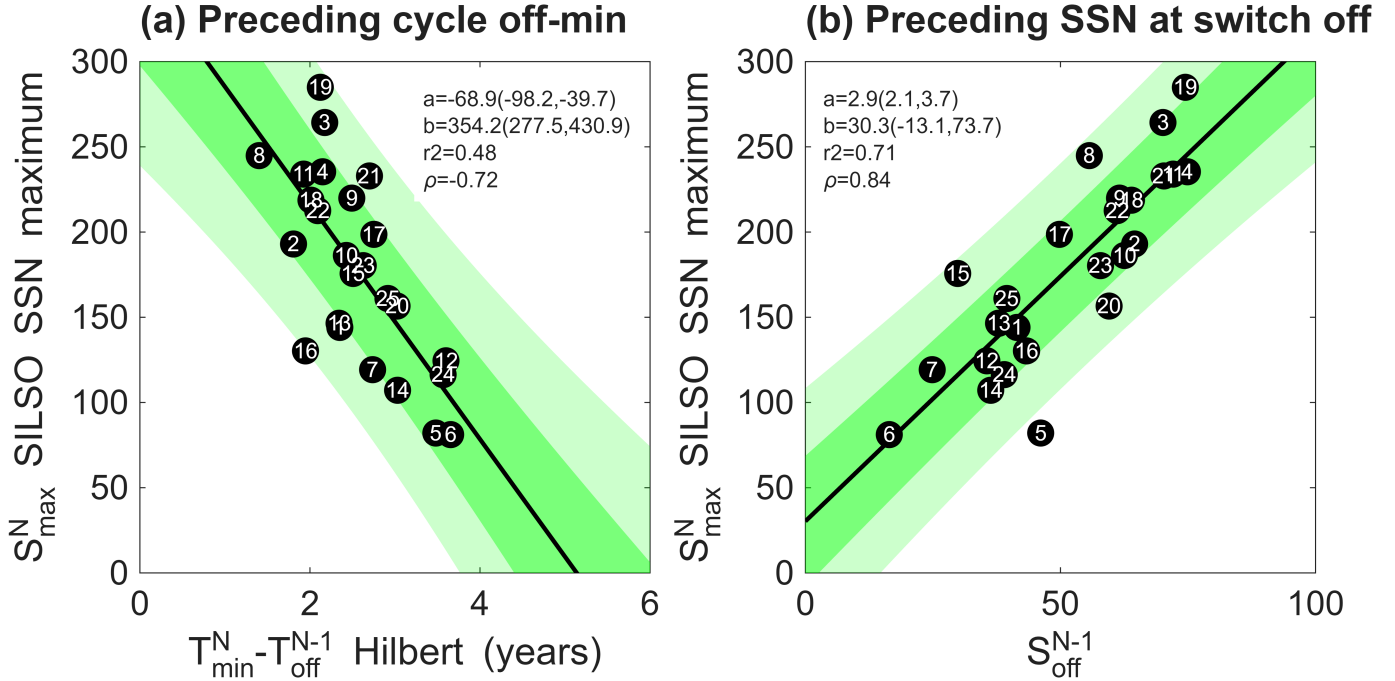


Figure 4. Correlation of the solar maximum SSN with preceding solar cycle declining phase. Linear regression (black lines) with its 68% and 95% confidence bounds (dark- and light-green shading) of each SSN solar maximum from SILSO plotted versus preceding cycle properties: (a) time interval between minimum and switch-off and (b) SSN at switch-off. Black circles indicate each cycle where the cycle number follows from Figure 1. The coefficients a , b with 95% confidence bounds for the linear regression of $y = ax + b$, the r^2 coefficient of determination, and the Pearson correlation coefficient are given in insets in each panel.

switch-off of cycle 25. The switch-off will have occurred when the majority of the AR centroids move to within 15° of the solar equator (blue lines in Figure 3), but this only provides an approximate timing. To obtain the switch-off timing from

Hilbert transform of the SSN will, at this stage in solar cycle 25, require forward projection of the SSN. This will also provide a projected value of the SSN at switch-off, giving a prediction for cycle 26 maximum.

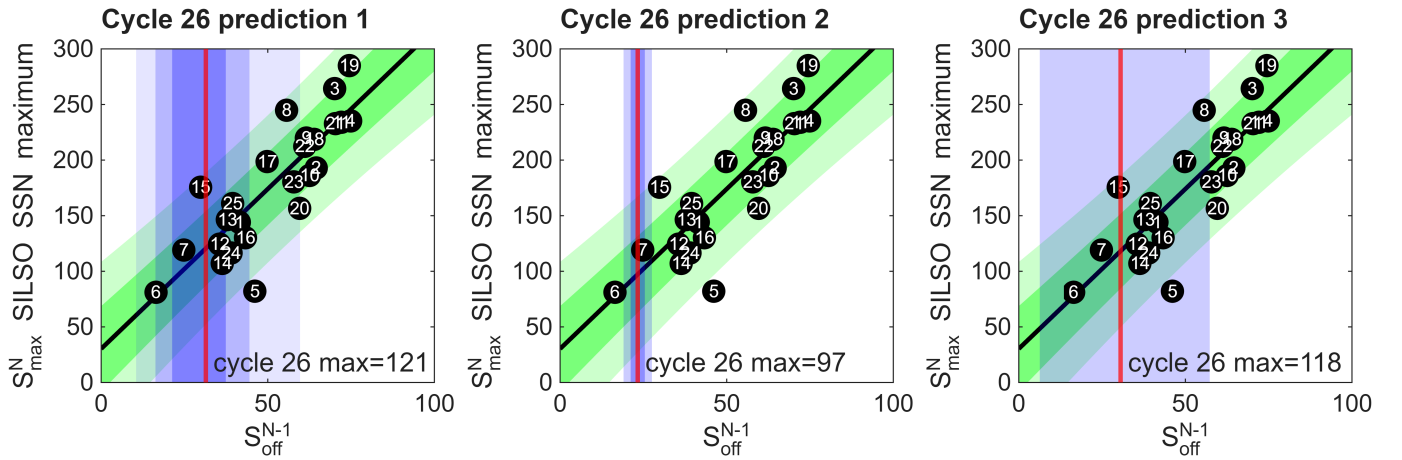


Figure 5. Prediction of cycle 26 maximum. Black numbered circles plot the SILSO SSN maxima versus the SSN at switch-off of the preceding cycle. The linear regression (black line) from Figure 4(b) is shown with its 68% and 95% confidence bounds (dark- and light-green shading). Vertical red lines indicate the cycle 25 switch-off SSN obtained from projecting cycle 25 forward using (1) NOAA (M. S. Miesch 2025), (2) (S. W. McIntosh et al. 2023), and (3) NMSC (K. O. Niehuss et al. 1996). Blue shading from dark to light plots the projection confidence bounds, prediction 1 [25, 50, 75]%, prediction 2 [68,95]%, and prediction 3 [95]%.

4. Hindcasting the Cycle Maxima

Figure 4 plots linear regressions of the SILSO SSN maximum of each cycle with properties of the declining phase of the preceding cycle derived from the switch-off time for cycles 1–25. Having determined each cycle switch-off time, there are two observations that can be considered: (a) the declining phase timescale, that is, the time between the switch-off and following minimum, and (b) the SSN at the switch-off. The Pearson correlation ρ and coefficient of determination r^2 capture both the strength of the linear correlation and the degree of scatter around the fitted line (Appendix C). The SSN at the switch-off (panel (b)) performs well, with $r^2 = 0.71$ and $\rho = 0.84$. This is our main result, that the SSN at switch-off of each cycle correlates with the maximum of next cycle. This provides a new predictor of upcoming solar maxima on a time horizon of about 7 yr, and it will be used here as the basis of a prediction for cycle 26 maximum. The declining phase timescale (panel (a)) also shows this correlation with a similar magnitude of $\rho = 0.72$ but smaller $r^2 = 0.48$. This is consistent with the finding of a link between the rate of decline of the solar dipole and the upcoming cycle maximum (P. Jaswal et al. 2024).

There is long-standing evidence that the maximum of the current cycle linearly correlates with the length of the preceding cycle (D. H. Hathaway et al. 1994, 2002). Linear regressions of cycle maximum upon preceding cycle length, for different estimates of the cycle length, are plotted in Appendix D (Figure 9). Overall, these regressions based on cycle length do not perform as well as the SSN at switch-off (Figure 4(b)). This is consistent with the stronger correlation that has been identified with the SSN 3 yr before, rather than at, the minimum (R. Cameron & M. Schüssler 2007).

A prediction for cycle 26 maximum with this range of uncertainties requires the 13 month smoothed SSN at the switch-off of current cycle 25, which (Figure 3) has not yet been reached. A current prediction then requires projections of the current cycle forward to the switch-off.

5. Predicting Cycle 26 Maximum

In order to predict the maximum of solar cycle 26, the monthly SSN is projected forward through the declining phase.

The three projections, which are in current use, are that of (1) NOAA (M. S. Miesch 2025), (2) McIntosh (2023) (S. W. McIntosh et al. 2023), and (3) NMSC (K. O. Niehuss et al. 1996; R. L. Holland et al. 1982) and are described in Appendix E. The SSN is first extended with these projections, then Hilbert transformed to forecast the occurrence time and the SSN at the switch-off of cycle 25. We can then read off predictions of the maximum of cycle 26 from each of the three forecasts of cycle 25 switch-off SSN, as shown in Figure 5.

Taken together, the three predictions are consistent with a weak-to-moderate cycle 26 maximum. The gradient of the linear regression of SILSO upcoming SSN maximum upon preceding cycle switch of SSN is ~ 3 , stretching the already broad (with the exception of projection 2) confidence bounds on the projections. The difficulty with projections with broad uncertainties is that they do not exclude either strong or weak cycles. These projections give (Appendix E, Figure 10) [date, SSN] for cycle 25 switch-off of (1) NOAA [2029.16,31], (2) S. W. McIntosh et al. (2023) [2028.88,23], and (3) NMSC [2029.15,31], that is, the switch-off is expected in about 2.5–3 yr. Once the switch-off has been crossed with sufficient margin, the cycle 26 maximum prediction relies solely on the observed SSN, with an uncertainty that will just be that due to the scatter of the regression, as shown in Figure 4(b). These confidence bounds will be sufficiently tight to discriminate between a weak, moderate, and strong cycle 26.

The SSN projections give predictions for the cycle 26 maximum that are consistent with each other and suggest a weak-to-moderate cycle 26 maximum, weaker than cycle 25. However, projections 1 and 2 are not independent as they are both based (Appendix E) on Hathaway’s model for the average shape of the solar cycle (D. H. Hathaway et al. 1994; D. H. Hathaway 2015), with different estimates of the parameters. Projection 3 uses a statistical method based on the average distribution of past cycle shapes. Observations suggest that strong cycles have a different shape to weak ones (R. M. Wilson et al. 1998). Potentially then, the projections, based on a single average form for the cycle shape, may not perform equally well in capturing the SSN at the switch-off for strong and weak cycles. This can be investigated using the timings from the solar cycle clock.

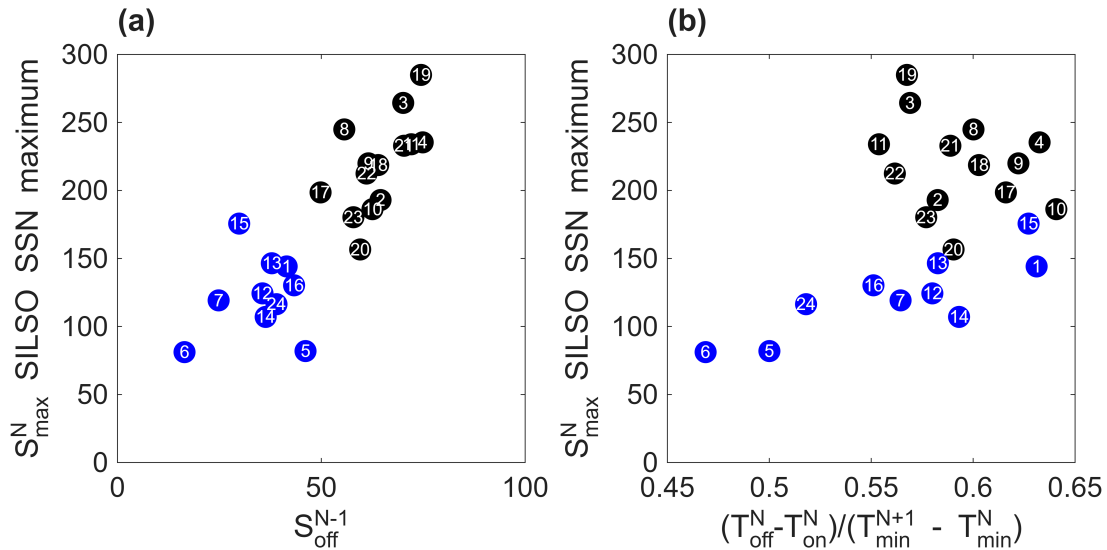


Figure 6. Duty cycle of the solar cycle. The SILSO solar cycle maximum is plotted versus a dimensionless duty cycle, a parameter for the cycle shape. The SILSO maximum is plotted versus (a) the preceding switch-off SSN as in Figure 4 and (b) the duty cycle, that is, the ratio of the duration of the active phase (switch-on to switch-off) divided by the cycle length between the next and the preceding minima. All timings are obtained from the Hilbert transform of the SSN (see the text). Panel (a) suggests that the solar cycles fall into two groupings colored blue and black on both panels.

6. Duty Cycle of the Solar Cycle

The duty cycle of a system is the ratio of the active period to the total period—it is a dimensionless parameter for the cycle shape. Here, the cycle active period, which contains the cycle maximum, is the time interval between the cycle switch-on and switch-off. The total period is the time interval between the minima before and after this maximum. The duty cycle for each of the Schwabe cycles 1–24 is shown in Figure 6. In the plot of the SILSO maxima against the preceding switch-off SSN (Figure 4(b)), the cycles show a tendency to form two populations—this is highlighted by repeating the plot in Figure 6(a) with color coding weak (blue) and strong (black) cycles. Figure 6(b) plots the SILSO maxima against the duty cycle with the same color coding. More active cycles that reach larger SSN maxima have a larger duty cycle, that is, they spend a larger fraction of the cycle in an active state. This in itself is relevant to the overall space weather impact at Earth since the most extreme events are confined to the active phase of the cycle (S. C. Chapman et al. 2020; S. C. Chapman & T. Dudok de Wit 2024). The plots also show a clear distinction between weak and strong cycles. For cycles with maxima SSN up to about 150, there is a roughly linear relationship between the SILSO maximum and the duty cycle. Cycles with maxima above 150 SSN exhibit a broad range of cycle maxima for a given duty cycle.

7. Discussion

SSN predictions on solar cycle timescales can be critical for mitigation of the effects of space weather on technological infrastructure at Earth, but are challenging as no two past cycles have the same amplitude or duration. The Hilbert transform of 13 month smoothed SSN since 1749 is used to identify the switch-off of the active phase of cycles 1–24. The SSN at the switch-off is shown to correlate with the following cycle maximum, forming the basis of a prediction with a ~ 7 yr time horizon. Since the current cycle 25 switch-off will not occur for at least another ~ 2.5 yr, a prediction for cycle 26

maximum requires projecting the current SSN forward to give a prediction of a weak-to-moderate cycle.

This prediction has large uncertainties that are driven by the uncertainties in the forward projection of the SSN. Once the switch-off of cycle 25 has occurred, an updated prediction with tighter uncertainty bounds will be possible; these uncertainties will be solely dictated by the scatter in the linear regression between the maximum SSN and the SSN at the switch-off of the preceding cycle seen in cycles 1–25. The uncertainties, in particular the upper bound, are as important as the central prediction, if not more so, for end users who typically need to plan for worst-case scenarios. This favors precursor predictors that have high values of both the coefficient of determination (r^2) and Pearson correlation (ρ), as is the case here.

The polar magnetic fields during the declining phase of the cycle “seed” the toroidal fields that drive sunspot activity of the next cycle (R. Cameron & M. Schüssler 2015). The switch-off identifies the specific phase of the cycle where this relationship should unambiguously be seen, both in models and in high-resolution observations. It suggests a physical explanation for the correlation found here between the cycle maximum and the preceding SSN at switch-off: the SSN at switch-off parameterizes the solar state at the onset of the declining phase. The switch-off is when solar eruptions change in character from coronal mass ejections to high-speed streams (S. C. Chapman & T. Dudok de Wit 2024), correlating both with the sunspot ARs moving to lower solar latitudes with reduced differential rotation and the switch-off of extreme space weather at earth (S. C. Chapman et al. 2020).

The question then arises as to whether more detailed, precise information about the solar state at the switch-off time, rather than just the SSN, would correlate more closely with the upcoming cycle amplitude. However, individual ARs with atypical properties can significantly perturb the development of the solar dynamo (M. Nagy et al. 2017) so that obtaining a tighter correlation may be challenging.

Finally, the shape of each Schwabe solar cycle is characterized by a dimensionless parameter—the duty cycle. The duty cycles for Schwabe cycles 1–24 suggest two modes

of solar cycle shape. Weak-to-moderate cycles with maximum $\lesssim 150$ have a roughly linear relationship between the cycle maximum and the duty cycle, whereas strong cycles have large scatter in this relationship. This may have implications for SSN projections that rely on the concept of an average solar cycle shape.

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Author Contributions

S.C.C. devised and performed the analysis and wrote the manuscript.

Data Availability

All data used in this study is freely available from the following sources (accessed on 2026 April 17).

SILSO Royal Observatory of Belgium, Brussels monthly total sunspot number version 2.0 from 1749: <http://www.sidc.be/silso/home>.

The dates of solar cycle maxima and minima are as determined from the smoothed sunspot number record by SILSO: <http://www.sidc.be/silso/cyclesmm>.

The active region areas' record reports the total sunspot area (in units of millionths of a hemisphere) found in 50 latitude bins distributed uniformly in sine(latitude) per Carrington rotation since 1874, available from http://solarcyclescience.com/AR_Database/bflydata.txt.

NOAA projection: <https://www.swpc.noaa.gov/products/predicted-sunspot-number-and-radio-flux>.

McIntosh (2023) projection: <https://helioforecast.space/solarcycle>.

NMSC projection: <https://www.nasa.gov/solar-cycle/progression-and-forecast/>.

Appendix A

Hilbert Transform of the SSN

The detailed procedure is as follows. The SILSO monthly SSN is first extended- with forward predictions, and then used to generate two signals: (i) 13 month smoothed boxcar window running mean (consistent with SILSO; F. Clette & L. Lefèvre 2015) and (ii) slow trend, a robust local linear regression that downweights outliers (“rlowess”) using a 40 yr window (this essentially obtains the Gleissberg cycle; S. C. Chapman et al. 2021). This slow trend is subtracted from the 13 month smoothed SSN to give a detrended signal, which importantly is zero crossing twice every Schwabe cycle. The detrended signal is then interpolated (cubic spline) to a daily time base to optimize mapping between specific Hilbert phases and times. The beginning of this signal is trimmed to give a whole number of cycles as input to the discrete Hilbert transform since it relies on an FFT; this trimmed signal is plotted in Figures 2(a), (b). The discrete analytic signal is obtained by a standard method (S. L. Marple 1999) that satisfies both

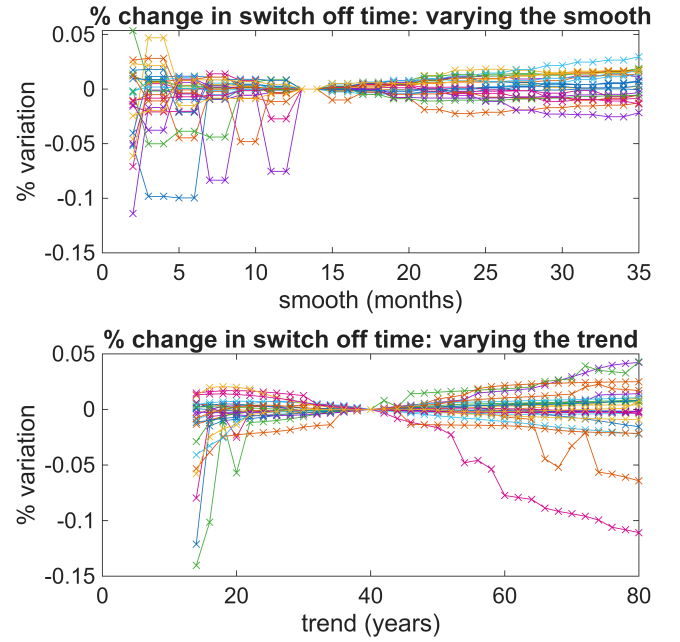


Figure 7. Variation in the switch-off timings as the SSN smooth parameters are varied. Colored lines/symbols (x) plot the percentage change in the switch-off time obtained for each solar cycle as the length of the window used to perform the (upper panel) smooth and (lower panel) subtracted trend. The percentage change is the difference from the value obtained for 13 month smooth and 40 yr trend.

invertability and orthogonality. The resulting phase $\phi(t)$ is plotted in Figure 2(c), which is inverted to obtain the switch-on and -off of each cycle. The stability of the switch-on and -off timings has been investigated (S. C. Chapman et al. 2020; S. W. McIntosh et al. 2020); Figure 7 plots the percentage change in the switch-off timings as the SSN smooth and trend windows are varied; for a reasonable range of parameters it is within $\sim \pm 0.2\%$.

Appendix B Extended Cycle

A simple model for the extended cycle is obtained by ordering the AR centroid latitudes with the SSN time series Hilbert analytic phase, as shown in Figure 8. The figure overplots for all cycles 1–25 the AR centroid latitude θ at the Hilbert phase ϕ of the midpoint of each Carrington rotation (black points). The AR centroids from multiple solar cycles fall around a single path in latitude and phase (S. C. Chapman & T. Dudok de Wit 2024), simplifying the (D. H. Hathaway 2011) detailed parametric fitting to the shape of the AR latitude time series. To model this trend, a linear least-squares regression is first obtained for the AR centroids for the most active half of the (normalized) cycle, from the switch-on at $+2\pi/5$ to half a cycle later at $2\pi/5 + \pi$ (overplotted blue points). This avoids fitting to lower latitudes where there is a zone of avoidance known as Spörer’s law of zones (D. H. Hathaway 2015). The fitted lines for each hemisphere are $\theta = a\phi + b$ with $a = -2.89, 3.36$ and $b = 23.38, -25.27$. These linear regression lines approximately intersect coordinates (Hilbert phase, latitude) of $((2n + 1)\pi, \pm \pi/12)$ and the switch-on $(2n\pi + 2\pi/5, 0)$, suggesting a simple extended cycle model.

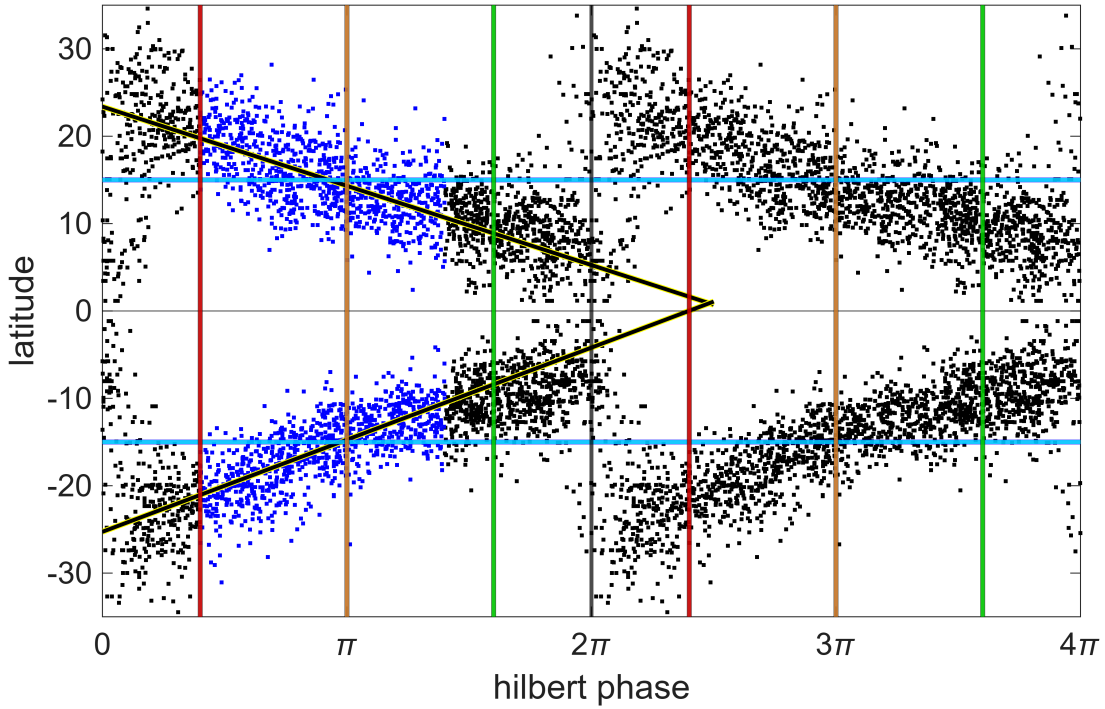


Figure 8. Simple model for the extended cycle. A simple model for the latitude of the extended cycle is obtained from the AR centroids. The figure plots the AR centroid latitude at the Hilbert phase at the midpoint of each Carrington rotation (black points). The model parameters are obtained from the linear least-squares regression for the most active half of the (normalized) cycle, from the switch-on at $+2\pi/5$ to half a cycle later at $2\pi/5 + \pi$ (overplotted blue points). The resulting fit (yellow highlighted black lines) is $\theta = a\phi + b$ with $a = -2.89, 3.36$ and $b = 23.38, -25.27$ for northern and southern hemispheres, respectively; this is extrapolated to model the extended cycle. The switch-on and -off are indicated by vertical red and green lines, respectively; the plot is centered on the average minimum, which is at zero (2π) phase (black line). The simple model extended cycle terminates (intersects zero latitude) approximately at the switch-on, $2\pi/5$ phase, and crosses 15° latitude at the π phase (horizontal blue line).

Appendix C Quantifying Goodness of Fit

Bisquare linear least-squares regression has been used throughout to obtain the linear regression of y upon x . The linear regression shown in Figures 4 and 5 is quantified in two ways. The r^2 coefficient of determination quantifies the extent to which the fitted linear relationship explains the data:

$$r^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}. \quad (\text{C1})$$

The Pearson correlation coefficient assesses the strength of a linear relationship:

$$\rho = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\left(\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{j=1}^n (y_j - \bar{y})^2\right)^{\frac{1}{2}}}. \quad (\text{C2})$$

Confidence intervals in Figures 4 and 5 are nonsimultaneous 68% and 95% observation bounds.

Appendix D Hindcasting the Next Solar Maximum with Current Solar Cycle Length

Figure 9 plots linear regression of solar maximum from SILSO versus preceding cycle length estimated from four different estimates of the duration of the preceding cycle: (a), (b) minimum to minimum, with the timings of the minima from SILSO, and from the Hilbert transform zero phase (c) switch-off to switch-off and (d) switch-on to switch-on.

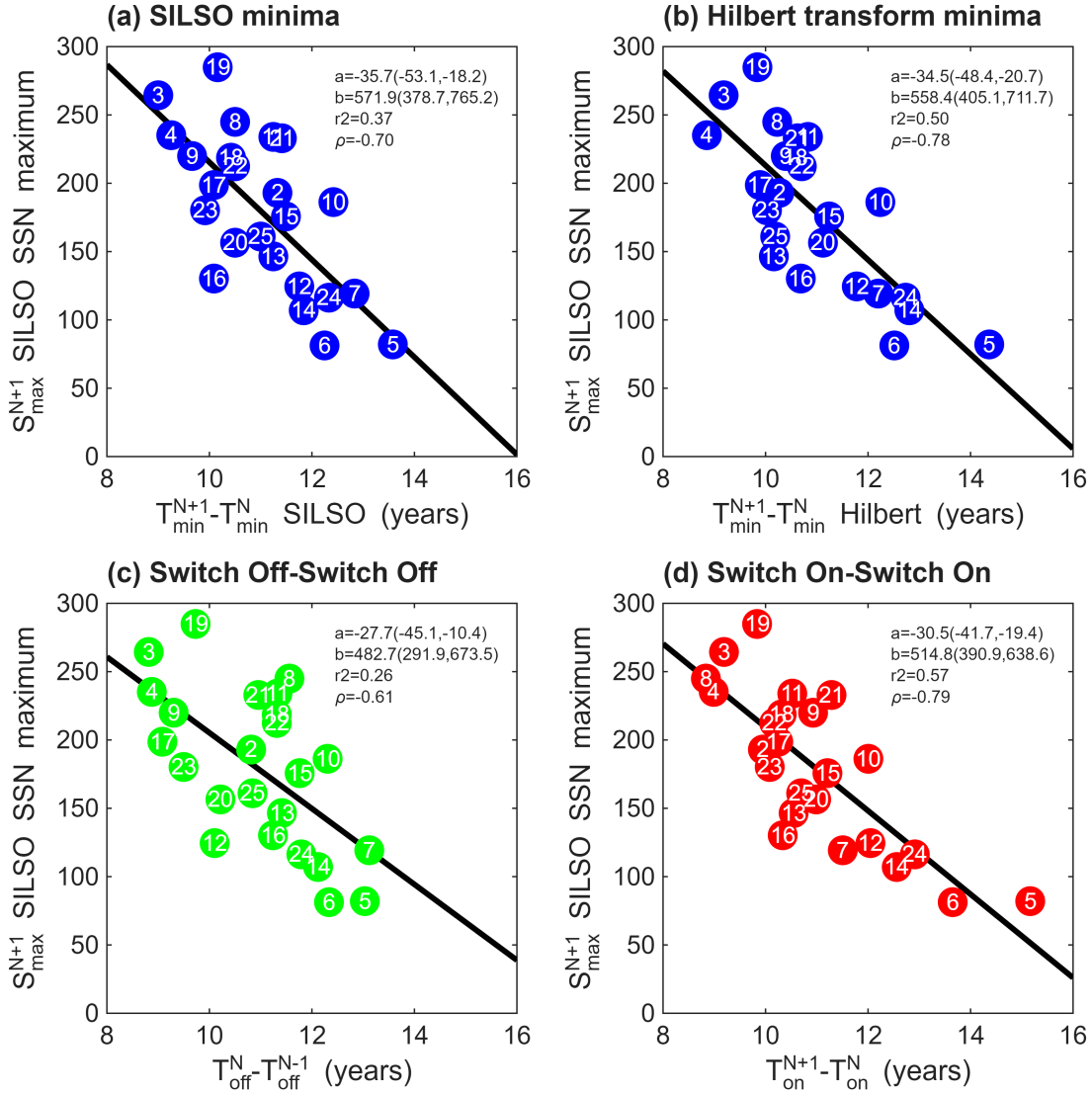


Figure 9. Correlation of next solar maximum with current solar cycle length. Linear regression (black lines) of solar maximum from SILSO plotted versus preceding cycle length estimated from (a), (b) minimum to minimum, with the timings of the minima from SILSO, and from the Hilbert transform zero phase (c) switch-off to switch-off and (d) switch-on to switch-on. Colored circles plot the values for each cycle where the cycle number follows from Figure 1. Linear regression coefficients with 95% confidence bounds, the r^2 coefficient of determination, and the Pearson correlation coefficient are given in insets in each panel.

Appendix E

Models to Project Forward through Cycle 25 Declining Phase

Figure 1 overplots the 13 month smoothed SSN record (blue) on the SSN projected forward with (1) NOAA (orange), (2) McIntosh (2023) (red), and (3) NMSC (pink) projections; Figure 10 provides a zoom-in of this plot showing cycles 23–25. These 13 month smooth and their 40 yr trends (light blue) are obtained for the monthly SSN record that has first been extended with monthly values of the projections, hence the smoothed, projected time series diverge from the smoothed SSN record in the vicinity of the end of the SSN record.

Both projections 1 and 2 rely upon the average shape of the solar cycle proposed by D. H. Hathaway et al. (1994) and D. H. Hathaway (2015):

$$f(t) = \frac{A(t - t_0)^3}{\exp(b - t_0^2)/b^2 - c}. \quad (\text{E1})$$

The NOAA projection (M. S. Miesch 2025) uses the empirical relationship (D. H. Hathaway et al. 1994) $b = 15.6 - 8.18A^{-\frac{1}{4}}$, with A and t_0 obtained from a fit to the observed monthly SSN, updated monthly. The McIntosh (2023) projection (S. W. McIntosh et al. 2023) uses $A = 365$, $b = 43.3$, $c = 0.71$, and t_0 of 2020 August 1. The NMSC projection (K. O. Niehuss et al. 1996) takes the empirical average shape of previous cycles as a starting point, then adjusts the current cycle prediction based on the observed deviation from the average, achieved by stretching the historical distribution (A. G. McNish & J. V. Lincoln 1949; R. L. Holland et al. 1982). NMSC currently report the distribution 5, 50, and 95 percentiles. The NOAA SSN projection provides monthly updated confidence bounds at 25%, 50%, and 75%. The McIntosh (2023) SSN projection gives a 68%, 1σ confidence bound on $A = \pm 33$ from which a 2σ 95% confidence level can be inferred, assuming a Gaussian distributed sample. The confidence limits on these projections are shown in Figure 5.

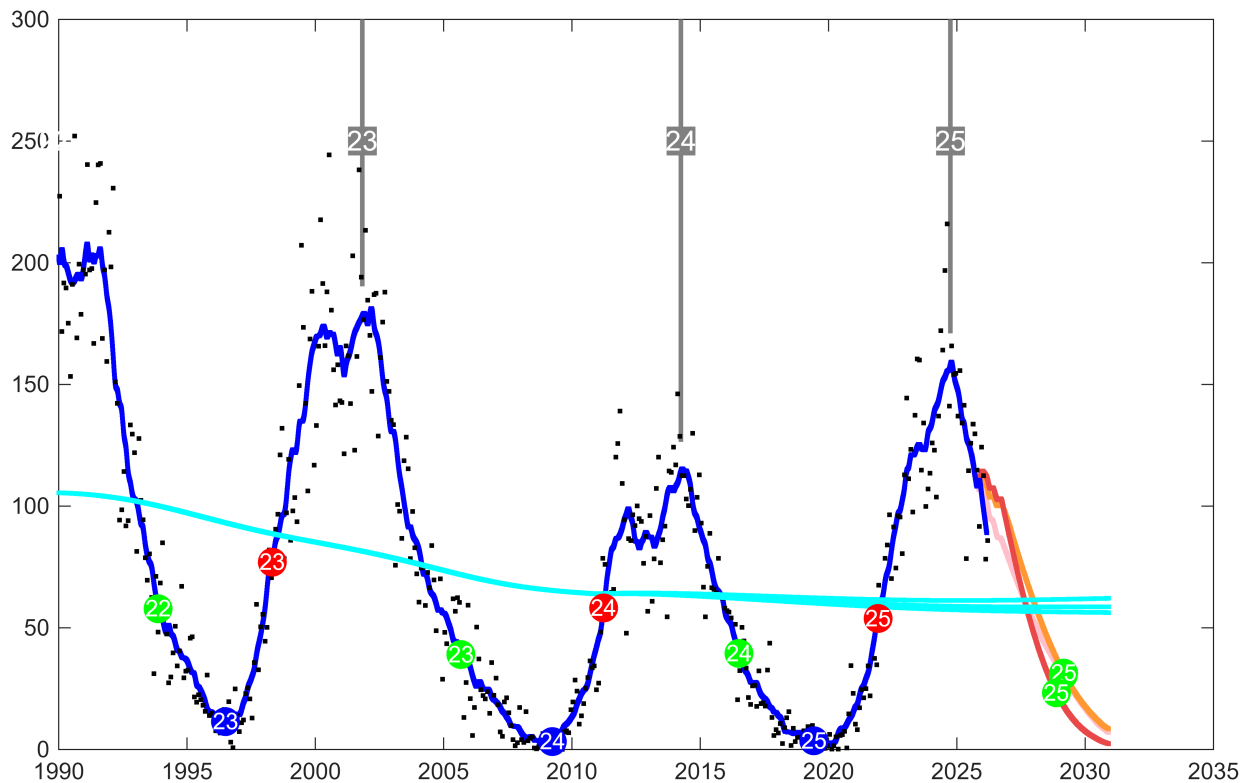


Figure 10. A zoom-in of Figure 1 showing the last few solar cycles. The monthly SSN (black dots) is overplotted with the 13 month smoothed SSN (blue), which is extended with Projection 1: NOAA (orange), Projection 2: (2023) (S. W. McIntosh et al. 2023) (red), and Projection 3: NMSC (K. O. Niehuss et al. 1996) (pink). The 40 yr trend including these projections is plotted (cyan). Each SILSO cycle maximum is indicated by gray vertical lines with the cycle number indicated. Colored numbered circles indicate each cycle minimum (blue), switch-off (green), and switch-on (red) at times determined by the Hilbert transform of projection-extended SSN. The numbering convention is that a cycle maximum follows a cycle minimum, so that the N th cycle minimum occurs first, followed by the N th cycle switch-on, then maximum, then N th cycle switch-off.

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